Chapter 1.2 Study Guide: Wireless and Mobile Networks

1. Overview

- Focuses on Wireless and Mobile Networks.
- Key objectives:
* Understand wireless network components and characteristics.
* Learn 802.11 (Wi-Fi), 4G, and 5G technologies.
* Grasp mobility management and Mobile IP.

2. Elements of a Wireless Network

- **Wireless hosts**: Laptops, smartphones, IoT devices.
- **Base station**: Connects wireless hosts to wired network (e.g., cell towers, Wi-Fi APs).
- **Wireless link**: Connects hosts to base stations; uses multiple access protocols.
- **Infrastructure mode**: Has base station connecting mobiles to Internet.
- **Ad hoc mode**: No base station; devices communicate directly (e.g., Bluetooth, MANET).

3. Wireless Link Characteristics

- Signal strength decreases over distance (path loss).
- Interference from other devices (2.4 GHz shared band).
- Multipath propagation: reflections cause delayed signals.
- **SNR (Signal-to-Noise Ratio)**:
* Higher SNR → lower BER (bit error rate).
* Adjust power or modulation based on SNR.
- **Hidden terminal problem**: Two nodes cannot detect each other but interfere at receiver.

4. CDMA (Code Division Multiple Access)

- Each user assigned a unique code (chipping sequence).
- All users share same frequency but transmit simultaneously.
- Minimizes interference if codes are orthogonal.
- Used in cellular networks.

5. IEEE 802.11 (Wi-Fi) Standards

Standard	Year	Max Rate	Range	Frequency
802.11b	1999	11 Mbps	30 m	2.4 GHz
802.11g	2003	54 Mbps	30 m	2.4 GHz
802.11n (WiFi 4)	2009	600 Mbps	70 m	2.4/5 GHz
802.11ac (WiFi 5)	2013	3.5 Gbps	70 m	5 GHz
802.11ax (WiFi 6)	2020	14 Gbps	70 m	2.4/5 GHz

- Uses **CSMA/CA** (Collision Avoidance).

6. IEEE 802.11 Architecture and Association

- **BSS (Basic Service Set)**: Access point + wireless hosts.
- Host scans for AP beacons (SSID, MAC) and associates with chosen AP.
- **Passive scanning**: Host listens for beacon frames.
- **Active scanning**: Host sends probe requests, APs respond.
- Authentication and DHCP follow association.

7. Medium Access Control (CSMA/CA)

- Sense channel before transmitting.
- No collision detection → uses ACKs and random backoff timers.
- **RTS/CTS mechanism**: Reserves channel to prevent hidden terminal collisions.

8. IEEE 802.11 Frame Structure

- Contains up to four MAC addresses (source, destination, AP, router).
- Includes sequence number, control fields, and CRC.
- Frame control bits identify type (RTS, CTS, ACK, data).

9. Advanced Capabilities in Wi-Fi

- **Rate adaptation**: Adjust transmission rate based on SNR.
- **Power management**: Device sleeps between beacon frames to save power.

10. Bluetooth (WPAN)

- Short-range (<10 m), 2.4 GHz band, up to 3 Mbps.
- **Piconet**: 1 master + up to 7 active slaves.
- Uses **FHSS (Frequency Hopping Spread Spectrum)** to minimize interference.
- **Scatternet**: Multiple piconets linked together.

11. 4G/5G Cellular Networks

- **4G (LTE)**:
* Based on 3GPP standards.
* All-IP network with up to 100s Mbps.
* Architecture components:
- eNodeB (base station)
- MME (Mobility Management Entity)
- S-GW (Serving Gateway)
- P-GW (Packet Data Network Gateway)
- HSS (Home Subscriber Service)
- **5G**:
* Goal: 10x speed, 10x lower latency.
* Uses millimeter-wave frequencies (FR1/FR2 bands).
* Requires dense deployment of small cells.

12. Mobility Management

- **Home Network**: Stores subscriber info (HSS).
- **Visited Network**: Provides service while roaming.
- **Mobility Approaches**:

* Indirect routing: Data passes through home network (triangle routing).
* Direct routing: Correspondent contacts visited network directly.
- **Mobile IP**:
* Early standard using tunnels for mobility (RFC 5944).
* Home Agent ↔ Foreign Agent concept.

13. Mobility in 4G Networks

- MME handles:
* Authentication with HSS.
* Handover between cells.
* Tunnel setup (GTP protocol).
- **Handover**: Device moves between base stations; tunnels updated.
- **Sleep modes**: Light (ms) and deep (seconds) to conserve battery.

14. Impact of Mobility on Higher Layers

- Wireless errors → packet loss, delay.
- TCP interprets loss as congestion → unnecessary slowdown.
- Real-time apps (VoIP) suffer from variable delay.

Exam Hotspots Summary

■ Understand Wi-Fi architecture, CSMA/CA, RTS/CTS.
■ Know 4G/5G structure and key components (MME, S-GW, P-GW, HSS).
■ Explain mobility (indirect vs direct routing, Mobile IP).
■ Recall Bluetooth structure and frequency range.
■ Describe wireless link issues: interference, SNR, hidden terminals.